Technical Research Report

Power Electronics and Control Systems

IGBTs performance analysis:

application to electrical drives.

Exploring different IGBT for an electrical drive.

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Abstract: This feasibility study investigates the potential adoption of the FF1200R17IP5P (Infineon) IGBT module as an alternative to the currently used DP650B1700 (Danfoss), FF650R17IE4 (Infineon), and FF650R17IE4DB2 (Infineon) in the 690 V EDrive platform. In parallel to a theoretical power losses evaluation a series of Double Pulse Tests (DPTs) were carried out to characterize and compare the switching behavior of the main module DP650B1700 (Danfoss series) and FF1200R17IP5P (new Infineon) under representative operating conditions. The evaluation also covers compatibility with both the legacy DB1.00 and the updated DB1.51 driver boards, in order to assess whether any hardware changes are required for a potential transition. The results and observations from this study aim to support the suitability of the FF1200R17IP5P as a replacement for the DP650B1700, FF650R17IE4DB2, and FF650R17IE4 modules.

Project Repository: repositories-documentation.git

Framework: MATLAB/Simulink/Simscape

Contents

0.1 Motivations: original design versus new performance requirements 6
0.1.1 New performance requirements 7
0.1.2 IGBTs behaviour under new performance requirements: theoretical calculation 8
0.1.3 IGBTs behaviour under new performance requirements: simulation results 13
0.1.3.1 Simulation results in tables 17
0.2 Double pulse tests 18
0.3 Measurement Results 19
0.3.1 Turn-OFF 19
0.3.2 Turn-ON 21
0.3.3 Summary of Test Results 24
0.4 Overall Results and Conclusions 26

List of Figures

1 Thermal behaviour comparison between IGBT DP650B1700 and FF1200R17IP5P at $I_{\text{load}}^{\text{inv}} = 370\,\text{A}$. 14
2 Thermal behaviour comparison between IGBT DP650B1700 and FF1200R17IP5P at $I_{\text{load}}^{\text{inv}} = 520\,\text{A}$. 15
3 Thermal behaviour comparison between IGBT FF650R17IE4 and FF1200R17IP5P at $I_{\text{load}}^{\text{inv}} = 370\,\text{A}$. 16
4 Thermal behaviour comparison between IGBT FF650R17IE4 and FF1200R17IP5P at $I_{\text{load}}^{\text{inv}} = 520\,\text{A}$. 17
5 Turn-OFF comparison: DP650B1700 with DB1.00 and DB1.51. 19
6 Turn-OFF comparison: FF1200R17IP5P with DB1.00 and DB1.51. 20
7 Turn-OFF comparison: DP650B1700 and FF1200R17IP5P with DB1.00. . 20
8 Turn-OFF comparison: DP650B1700 and FF1200R17IP5P with DB1.51. . 21
9 Turn-ON comparison: DP650B1700 with DB1.00 and DB1.51. 21
10 Turn-ON comparison: FF1200R17IP5P with DB1.00 and DB1.51. 22
11 Turn-ON comparison: DP650B1700 and FF1200R17IP5P with DB1.00. . . 22
12 Turn-ON comparison: DP650B1700 and FF1200R17IP5P with DB1.51. . . 23

List of Tables

1 **Original Design Inverter** — Power losses and maximum junction temperature at $I_{\text{load}}^{\text{inv}} = 370\,\text{A}$, $f_{\text{pwm}}^{\text{inv}} = 2.5\,\text{kHz}$, $T_{\text{hs}}^{\text{inv}} = 65\,^{\circ}\text{C}$, $\cos(\varphi) = -0.8$, with DB1.00. 6

2 **Original Design AFE** — Power losses and maximum junction temperature at $I_{\text{load}}^{\text{inv}} = 270\,\text{A}$, $f_{\text{pwm}}^{\text{inv}} = 4\,\text{kHz}$, $T_{\text{hs}}^{\text{inv}} = 65\,^{\circ}\text{C}$, $\cos(\varphi) = 1$, with DB1.00. 7

3 **Case Scenario 1** — Power losses and maximum junction temperature at $I_{\text{load}}^{\text{inv}} = 370\,\text{A}$, $f_{\text{pwm}}^{\text{inv}} = 4\,\text{kHz}$, $T_{\text{hs}}^{\text{inv}} = 65\,^{\circ}\text{C}$, $\cos(\varphi) = 0.8$, with DB1.00. . . . 8

4 **Case Scenario 2** — Power losses and maximum junction temperature at $I_{\text{load}}^{\text{inv}} = 520\,\text{A}$, $f_{\text{pwm}}^{\text{inv}} = 4\,\text{kHz}$, $T_{\text{hs}}^{\text{inv}} = 65\,^{\circ}\text{C}$, $\cos(\varphi) = 0.8$, with DB1.00. . . . 9

5 **Case Scenario 3** — Power losses and maximum junction temperature at $I_{\text{load}}^{\text{inv}} = 370\,\text{A}$, $f_{\text{pwm}}^{\text{inv}} = 4\,\text{kHz}$, $T_{\text{hs}}^{\text{inv}} = 65\,^{\circ}\text{C}$, $\cos(\varphi) = 0.8$, with DB1.51. . . . 10

6 **Case Scenario 4** — Power losses and maximum junction temperature at $I_{\text{load}}^{\text{inv}} = 520\,\text{A}$, $f_{\text{pwm}}^{\text{inv}} = 4\,\text{kHz}$, $T_{\text{hs}}^{\text{inv}} = 65\,^{\circ}\text{C}$, $\cos(\varphi) = 0.8$, with DB1.51. . . . 10

7 **Case Scenario 5** — Power losses and maximum junction temperature at $I_{\text{load}}^{\text{inv}} = 370\,\text{A}$, $f_{\text{pwm}}^{\text{inv}} = 4\,\text{kHz}$, $T_{\text{hs}}^{\text{inv}} = 65\,^{\circ}\text{C}$, $\cos(\varphi) = -0.8$, with DB1.00. . . 11

8 **Case Scenario 6** — Power losses and maximum junction temperature at $I_{\text{load}}^{\text{inv}} = 520\,\text{A}$, $f_{\text{pwm}}^{\text{inv}} = 4\,\text{kHz}$, $T_{\text{hs}}^{\text{inv}} = 65\,^{\circ}\text{C}$, $\cos(\varphi) = -0.8$, with DB1.00. . . 12

9 **Case Scenario 7** — Power losses and maximum junction temperature at $I_{\text{load}}^{\text{inv}} = 370\,\text{A}$, $f_{\text{pwm}}^{\text{inv}} = 4\,\text{kHz}$, $T_{\text{hs}}^{\text{inv}} = 65\,^{\circ}\text{C}$, $\cos(\varphi) = -0.8$, with DB1.51. . . 12

10 **Case Scenario 8** — Power losses and maximum junction temperature at $I_{\text{load}}^{\text{inv}} = 520\,\text{A}$, $f_{\text{pwm}}^{\text{inv}} = 4\,\text{kHz}$, $T_{\text{hs}}^{\text{inv}} = 65\,^{\circ}\text{C}$, $\cos(\varphi) = -0.8$, with DB1.51. . . 13

11 **Simulation results case scenario 1** — Power losses and maximum junction temperature at $I_{\text{load}}^{\text{inv}} = 370\,\text{A}$, $f_{\text{pwm}}^{\text{inv}} = 4\,\text{kHz}$, $T_{\text{hs}}^{\text{inv}} = 65\,^{\circ}\text{C}$, $\cos(\varphi) = 0.8$. 17

12 **Simulation results case scenario 2** — Power losses and maximum junction temperature at $I_{\text{load}}^{\text{inv}} = 520\,\text{A}$, $f_{\text{pwm}}^{\text{inv}} = 4\,\text{kHz}$, $T_{\text{hs}}^{\text{inv}} = 65\,^{\circ}\text{C}$, $\cos(\varphi) = 0.8$ 18

13 Summary of DPT results. 24

14 Reference E_{on} and E_{off} for the DPTs. 25

15 Measured E_{on} and E_{off} with DB1.51. 25

16 Measured E_{on} and E_{off} with DB1.00. 25

Nomenclature

- **AFE**: AC/DC three phase active front end;
- **Inverter**: DC/AC three phase inverter for PSM and IM electrical machines;
- **DPT**: Double Pulse Test;
- **FF1200R17IP5P**: Infineon module, 1200 A / 1700 V;
  - maximum junction temperature $T_{\mathrm{j}}^{\max} = 175\,^{\circ}\mathrm{C}$;
  - **new module under evaluation**;
- **DP650B1700**: Danfoss module, 650 A / 1700 V;
  - maximum junction temperature $T_{\mathrm{j}}^{\max} = 150\,^{\circ}\mathrm{C}$;
  - currently employed in 690 V EDrive applications;
- **FF650R17IE4**: Infineon module, 650 A / 1700 V;
  - maximum junction temperature $T_{\mathrm{j}}^{\max} = 150\,^{\circ}\mathrm{C}$;
  - currently employed in 690 V EDrive applications;
- **FF650R17IE4DB2**: Infineon module, 650 A / 1700 V;
  - maximum junction temperature $T_{\mathrm{j}}^{\max} = 150\,^{\circ}\mathrm{C}$;
  - currently employed in 690 V EDrive applications;
- **DB1.00**: driver board currently employed in series production;
- **DB1.51**: **new driver board under evaluation**;
- E_{on} [mJ]: turn-on switching energy loss;
- E_{off} [mJ]: turn-off switching energy loss;
- I_{rr} [A]: reverse recovery current at turn-on;
- Δu_{ce} [V]: collector-emitter voltage overshoot at turn-off;
- $I_{\mathrm{nom}}^{\mathrm{inv}}$ [A]: nominal inverter current (RMS);
- $I_{\mathrm{max}}^{\mathrm{inv}}$ [A]: maximum overload inverter current (RMS);
- $I_{\mathrm{load}}^{\mathrm{inv}}$ [A]: inverter load current used for power loss calculation (RMS);
- $f_{\mathrm{pwm}}^{\mathrm{inv}}$ [Hz]: inverter switching frequency;
- $\cos(\varphi)$ [−]: inverter load power factor;
- $T_{\mathrm{j}}^{\mathrm{Q}}$ [°C]: IGBT junction temperature;
- $T_{\mathrm{j}}^{\mathrm{D}}$ [°C]: diode junction temperature;



- $P_{\mathrm{c}}^{\mathrm{Q}}$ [W]: IGBT conduction power loss;
- $P_{\mathrm{c}}^{\mathrm{D}}$ [W]: diode conduction power loss;
- $P_{\mathrm{sw}}^{\mathrm{Q}}$ [W]: IGBT switching power loss;
- $P_{\mathrm{sw}}^{\mathrm{D}}$ [W]: diode switching power loss;
- $\Delta T_{\mathrm{j}}^{\mathrm{Q}}$ [°C]: IGBT junction temperature ripple;
- $\Delta T_{\mathrm{j}}^{\mathrm{D}}$ [°C]: diode junction temperature ripple;
- $R_{\mathrm{g}}^{\mathrm{on}}$ [Ω]: turn-on gate resistance;
- $R_{\mathrm{g}}^{\mathrm{off}}$ [Ω]: turn-off gate resistance;
- $T_{\mathrm{hs}}^{\mathrm{inv}}$ [°C]: heat-sink temperature below the inverter IGBTs.

0.1 Motivations: original design versus new performance requirements

The EDrive 690 V has been designed primarily for wind applications with the following nominal specifications:

- $I_{\text{nom}}^{\text{inv}} = 370\,\text{A}$;
- $f_{\text{pwm}}^{\text{inv}} = 2.5\,\text{kHz}$.

Moreover, in wind applications, overloads are always related to wind speed (wind gusts), which are reflected on the AFE side. Therefore, the quantity $I_{\text{nom}}^{\text{inv}}$ can also be assumed as the maximum inverter current, and the typical overload capability of 140% of electrical drives here decays.

Inverter Design Criteria

- $I_{\text{load}}^{\text{inv}} = 370\,\text{A}$, 100% of the nominal inverter current;
- $f_{\text{pwm}}^{\text{inv}} = 2.5\,\text{kHz}$;
- $T_{\text{hs}}^{\text{inv}} = 65\,^{\circ}\text{C}$;
- $\cos(\varphi) = -0.8$, generator mode;
- $f_{\text{load}} = 15\,\text{Hz}$, load frequency (critical parameter for maximum junction temperature evaluation);
- driver board DB1.00.

Table 1: **Original Design Inverter** — Power losses and maximum junction temperature at $I_{\text{load}}^{\text{inv}} = 370\,\text{A}$, $f_{\text{pwm}}^{\text{inv}} = 2.5\,\text{kHz}$, $T_{\text{hs}}^{\text{inv}} = 65\,^{\circ}\text{C}$, $\cos(\varphi) = -0.8$, with DB1.00.

	T_{j}^{Q}	T_{j}^{D}	P_{c}^{Q}	P_{c}^{D}	P_{sw}^{Q}	P_{sw}^{D}	ΔT_{j}
FF650R17IE4DB2	95 °C	102 °C	71 W	181 W	337 W	127 W	19 °C

AFE Design Criteria

- $I_{\text{load}}^{\text{afe}} = 270\,\text{A}$, 135% of the nominal afe current;
- $f_{\text{pwm}}^{\text{afe}} = 4\,\text{kHz}$;, higher switching frequency for power quality constraints;

- $T_{\mathrm{hs}}^{\mathrm{afe}} = 65\,^{\circ}\mathrm{C}$;
- $\cos(\varphi) = 1$, worst case scenario in terms of power losses;
- $f_{\mathrm{grid}} = 50\,\mathrm{Hz}$, grid frequency;
- driver board DB1.00.

Table 2: **Original Design AFE** — Power losses and maximum junction temperature at $I_{\mathrm{load}}^{\mathrm{inv}} = 270\,\mathrm{A}$, $f_{\mathrm{pwm}}^{\mathrm{inv}} = 4\,\mathrm{kHz}$, $T_{\mathrm{hs}}^{\mathrm{inv}} = 65\,^{\circ}\mathrm{C}$, $\cos(\varphi) = 1$, with DB1.00.

	$T_{\mathrm{j}}^{\mathrm{Q}}$	$T_{\mathrm{j}}^{\mathrm{D}}$	$P_{\mathrm{c}}^{\mathrm{Q}}$	$P_{\mathrm{c}}^{\mathrm{D}}$	$P_{\mathrm{sw}}^{\mathrm{Q}}$	$P_{\mathrm{sw}}^{\mathrm{D}}$	ΔT_{j}
FF650R17IE4	97 °C	84 °C	163 W	24 W	397 W	139 W	6 °C

0.1.1 New performance requirements

Remark: In recent years, the increasing adoption of the LD690V in electrical drive applications has highlighted the need to raise the inverter overload current to $I_{\mathrm{max}}^{\mathrm{inv}} = 1.4 \times I_{\mathrm{nom}}^{\mathrm{inv}}$.

Remark: The use of hard parallelization requires operating at higher switching frequencies, bringing the worst-case scenario to $f_{\mathrm{pwm}}^{\mathrm{inv}} = 4\,\mathrm{kHz}$.

Remark: When $\cos(\varphi) > 0$, the inverter operates as a motor drive (first quadrant operation), which is the typical configuration for drive applications. In this condition, the IGBTs are more heavily loaded.

Remark: When $\cos(\varphi) < 0$, the inverter operates as a generator (second quadrant operation), which is the typical configuration for wind applications. In this condition, the diodes are more heavily loaded.

Remark: A general design rule requires the following condition to be satisfied:

$$T_{\mathrm{j}}^{\mathrm{design}}(t)\Big|_{\mathrm{max}} \leq T_{\mathrm{j}}^{\mathrm{max}} - 50\,^{\circ}\mathrm{C}. \tag{0.1.1}$$

The requirements outlined above highlight the need for a new IGBT module for the LD690V, which led to the selection of the FF1200R17IP5P.

0.1.2 IGBTs behaviour under new performance requirements: theoretical calculation

This subsection evaluates the power losses as well as the thermal behaviour of different IGBT modules under various load conditions, providing a comprehensive baseline for further investigations.

Analysis involve eight case scenario of load conditions, as follows.

Case Scenario 1

- $I_{\text{load}}^{\text{inv}} = 370\,\text{A}$, 100% of the nominal inverter current;
- $f_{\text{pwm}}^{\text{inv}} = 4\,\text{kHz}$;
- $T_{\text{hs}}^{\text{inv}} = 65\,^{\circ}\text{C}$;
- $\cos(\varphi) = 0.8$, drive mode;
- $f_{\text{load}} = 15\,\text{Hz}$, load frequency (critical parameter for maximum junction temperature evaluation);
- driver board DB1.00.

Table 3: **Case Scenario 1** — Power losses and maximum junction temperature at $I_{\text{load}}^{\text{inv}} = 370\,\text{A}$, $f_{\text{pwm}}^{\text{inv}} = 4\,\text{kHz}$, $T_{\text{hs}}^{\text{inv}} = 65\,^{\circ}\text{C}$, $\cos(\varphi) = 0.8$, with DB1.00.

	T_{j}^{Q}	T_{j}^{D}	P_{c}^{Q}	P_{c}^{D}	P_{sw}^{Q}	P_{sw}^{D}	ΔT_{j}
FF650R17IE4	120 °C	97 °C	242 W	54 W	576 W	175 W	25 °C
FF650R17IE4DB2	130 °C	95 °C	245 W	50 W	598 W	196 W	32 °C
FF1200R17IP5P	110 °C	83 °C	172 W	42 W	705 W	170 W	18 °C

Case Scenario 2

- $I_{\text{load}}^{\text{inv}} = 520\,\text{A}$, 140% of the nominal inverter current;
- $f_{\text{pwm}}^{\text{inv}} = 4\,\text{kHz}$;
- $T_{\text{hs}}^{\text{inv}} = 65\,^{\circ}\text{C}$;



– $\cos(\varphi) = 0.8$, drive mode;

– $f_{\text{load}} = 15\,\text{Hz}$, load frequency (critical parameter for maximum junction temperature evaluation);

– driver board DB1.00.

Table 4: **Case Scenario 2** — Power losses and maximum junction temperature at $I_{\text{load}}^{\text{inv}} = 520\,\text{A}$, $f_{\text{pwm}}^{\text{inv}} = 4\,\text{kHz}$, $T_{\text{hs}}^{\text{inv}} = 65\,^{\circ}\text{C}$, $\cos(\varphi) = 0.8$, with DB1.00.

	T_{j}^{Q}	T_{j}^{D}	P_{c}^{Q}	P_{c}^{D}	P_{sw}^{Q}	P_{sw}^{D}	ΔT_{j}
FF650R17IE4	156 °C	108 °C	420 W	85 W	911 W	220 W	42 °C
FF650R17IE4DB2	170 °C	103 °C	430 W	77 W	950 W	236 W	50 °C
FF1200R17IP5P	131 °C	89 °C	280 W	64 W	1000 W	220 W	27 °C

Case Scenario 3

– $I_{\text{load}}^{\text{inv}} = 370\,\text{A}$, 100% of the nominal inverter current;

– $f_{\text{pwm}}^{\text{inv}} = 4\,\text{kHz}$;

– $T_{\text{hs}}^{\text{inv}} = 65\,^{\circ}\text{C}$;

– $\cos(\varphi) = 0.8$, drive mode;

– $f_{\text{load}} = 15\,\text{Hz}$, load frequency (critical parameter for maximum junction temperature evaluation);

– driver board DB1.51 (new driver board).

Table 5: **Case Scenario 3** — Power losses and maximum junction temperature at $I_{\text{load}}^{\text{inv}} = 370\,\text{A}$, $f_{\text{pwm}}^{\text{inv}} = 4\,\text{kHz}$, $T_{\text{hs}}^{\text{inv}} = 65\,^{\circ}\text{C}$, $\cos(\varphi) = 0.8$, with DB1.51.

	T_{j}^{Q}	T_{j}^{D}	P_{c}^{Q}	P_{c}^{D}	P_{sw}^{Q}	P_{sw}^{D}	ΔT_{j}
FF650R17IE4	121 °C	98 °C	242 W	54 W	572 W	175 W	25 °C
FF650R17IE4DB2	128 °C	95 °C	245 W	49 W	585 W	198 W	30 °C
FF1200R17IP5P	107 °C	84 °C	172 W	42 W	644 W	171 W	18 °C

Case Scenario 4

- $I_{\text{load}}^{\text{inv}} = 520\,\text{A}$, 140% of the nominal inverter current;
- $f_{\text{pwm}}^{\text{inv}} = 4\,\text{kHz}$;
- $T_{\text{hs}}^{\text{inv}} = 65\,^{\circ}\text{C}$;
- $\cos(\varphi) = 0.8$, drive mode;
- $f_{\text{load}} = 15\,\text{Hz}$, load frequency (critical parameter for maximum junction temperature evaluation);
- driver board DB1.51 (new driver board).

Table 6: **Case Scenario 4** — Power losses and maximum junction temperature at $I_{\text{load}}^{\text{inv}} = 520\,\text{A}$, $f_{\text{pwm}}^{\text{inv}} = 4\,\text{kHz}$, $T_{\text{hs}}^{\text{inv}} = 65\,^{\circ}\text{C}$, $\cos(\varphi) = 0.8$, with DB1.51.

	T_{j}^{Q}	T_{j}^{D}	P_{c}^{Q}	P_{c}^{D}	P_{sw}^{Q}	P_{sw}^{D}	ΔT_{j}
FF650R17IE4	156 °C	110 °C	421 W	86 W	905 W	220 W	45 °C
FF650R17IE4DB2	169 °C	103 °C	431 W	77 W	927 W	240 W	50 °C
FF1200R17IP5P	127 °C	90 °C	280 W	65 W	913 W	222 W	25 °C

Case Scenario 5

- $I_{\text{load}}^{\text{inv}} = 370\,\text{A}$, 100% of the nominal inverter current;
- $f_{\text{pwm}}^{\text{inv}} = 4\,\text{kHz}$;
- $T_{\text{hs}}^{\text{inv}} = 65\,^{\circ}\text{C}$;
- $\cos(\varphi) = -0.8$, generator mode;
- $f_{\text{load}} = 15\,\text{Hz}$, load frequency (critical parameter for maximum junction temperature evaluation);
- driver board DB1.00.

Table 7: **Case Scenario 5** — Power losses and maximum junction temperature at $I_{\text{load}}^{\text{inv}} = 370\,\text{A}$, $f_{\text{pwm}}^{\text{inv}} = 4\,\text{kHz}$, $T_{\text{hs}}^{\text{inv}} = 65\,^{\circ}\text{C}$, $\cos(\varphi) = -0.8$, with DB1.00.

	T_{j}^{Q}	T_{j}^{D}	P_{c}^{Q}	P_{c}^{D}	P_{sw}^{Q}	P_{sw}^{D}	ΔT_{j}
FF650R17IE4	107 °C	124 °C	62 W	210 W	555 W	220 W	30 °C
FF650R17IE4DB2	112 °C	114 °C	62 W	189 W	568 W	215 W	25 °C
FF1200R17IP5P	102 °C	94 °C	45 W	159 W	688 W	179 W	13 °C

Case Scenario 6

- $I_{\text{load}}^{\text{inv}} = 520\,\text{A}$, 140% of the nominal inverter current;
- $f_{\text{pwm}}^{\text{inv}} = 4\,\text{kHz}$;
- $T_{\text{hs}}^{\text{inv}} = 65\,^{\circ}\text{C}$;
- $\cos(\varphi) = -0.8$, generator mode;
- $f_{\text{load}} = 15\,\text{Hz}$, load frequency (critical parameter for maximum junction temperature evaluation);
- driver board DB1.00.

Table 8: **Case Scenario 6** — Power losses and maximum junction temperature at $I_{\text{load}}^{\text{inv}} = 520\,\text{A}$, $f_{\text{pwm}}^{\text{inv}} = 4\,\text{kHz}$, $T_{\text{hs}}^{\text{inv}} = 65\,^{\circ}\text{C}$, $\cos(\varphi) = -0.8$, with DB1.00.

	T_{j}^{Q}	T_{j}^{D}	P_{c}^{Q}	P_{c}^{D}	P_{sw}^{Q}	P_{sw}^{D}	ΔT_{j}
FF650R17IE4	130 °C	155 °C	100 W	343 W	841 W	283 W	50 °C
FF650R17IE4DB2	137 °C	134 °C	104 W	299 W	862 W	277 W	35 °C
FF1200R17IP5P	118 °C	107 °C	72 W	247 W	964 W	240 W	32 °C

Case Scenario 7

- $I_{\text{load}}^{\text{inv}} = 370\,\text{A}$, 100% of the nominal inverter current;
- $f_{\text{pwm}}^{\text{inv}} = 4\,\text{kHz}$;
- $T_{\text{hs}}^{\text{inv}} = 65\,^{\circ}\text{C}$;
- $\cos(\varphi) = -0.8$, generator mode;
- $f_{\text{load}} = 15\,\text{Hz}$, load frequency (critical parameter for maximum junction temperature evaluation);
- driver board DB1.51 (new driver board).

Table 9: **Case Scenario 7** — Power losses and maximum junction temperature at $I_{\text{load}}^{\text{inv}} = 370\,\text{A}$, $f_{\text{pwm}}^{\text{inv}} = 4\,\text{kHz}$, $T_{\text{hs}}^{\text{inv}} = 65\,^{\circ}\text{C}$, $\cos(\varphi) = -0.8$, with DB1.51.

	T_{j}^{Q}	T_{j}^{D}	P_{c}^{Q}	P_{c}^{D}	P_{sw}^{Q}	P_{sw}^{D}	ΔT_{j}
FF650R17IE4	106 °C	124 °C	62 W	210 W	548 W	200 W	22.5 °C
FF650R17IE4DB2	111 °C	114 °C	62 W	189 W	557 W	219 W	22.5 °C
FF1200R17IP5P	100 °C	94 °C	45 W	159 W	629 W	181 W	12 °C

Case Scenario 8



- $I_{\text{load}}^{\text{inv}} = 520\,\text{A}$, 140% of the nominal inverter current;
- $f_{\text{pwm}}^{\text{inv}} = 4\,\text{kHz}$;
- $T_{\text{hs}}^{\text{inv}} = 65\,^{\circ}\text{C}$;
- $\cos(\varphi) = -0.8$, generator mode;
- $f_{\text{load}} = 15\,\text{Hz}$, load frequency (critical parameter for maximum junction temperature evaluation);
- driver board DB1.51 (new driver board).

Table 10: **Case Scenario 8** — Power losses and maximum junction temperature at $I_{\text{load}}^{\text{inv}} = 520\,\text{A}$, $f_{\text{pwm}}^{\text{inv}} = 4\,\text{kHz}$, $T_{\text{hs}}^{\text{inv}} = 65\,^{\circ}\text{C}$, $\cos(\varphi) = -0.8$, with DB1.51.

	T_{j}^{Q}	T_{j}^{D}	P_{c}^{Q}	P_{c}^{D}	P_{sw}^{Q}	P_{sw}^{D}	ΔT_{j}
FF650R17IE4	129 °C	155 °C	103 W	343 W	836 W	283 W	50 °C
FF650R17IE4DB2	136 °C	135 °C	104 W	299 W	843 W	282 W	35 °C
FF1200R17IP5P	114 °C	107 °C	72 W	247 W	878 W	243 W	30 °C

Data results outlined along the above tables indicates that the FF1200R17IP5P is a suitable candidate for the new performance requirements, here reported:

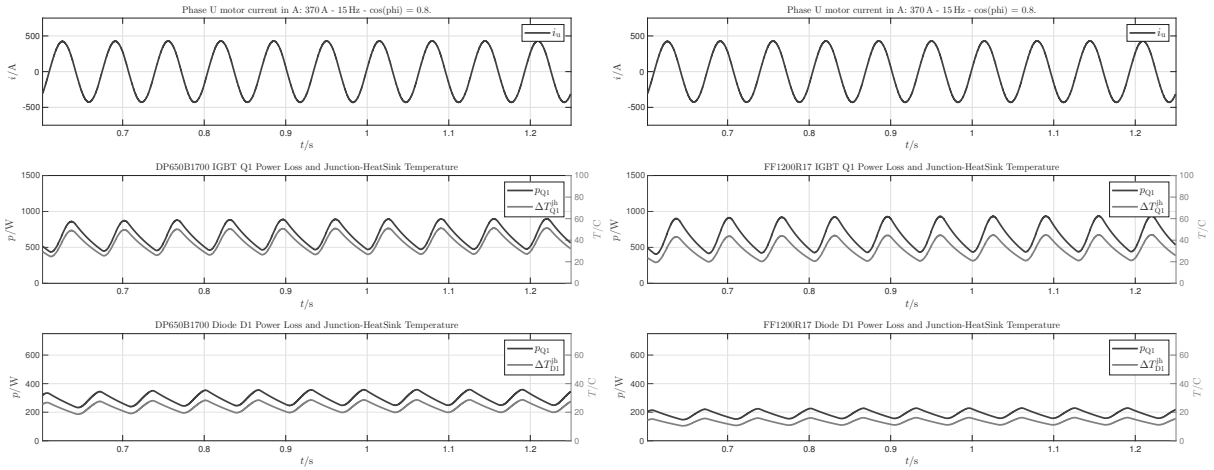
- $I_{\text{load}}^{\text{inv}} = 520\,\text{A}$, 140% of the nominal inverter current;
- $f_{\text{pwm}}^{\text{inv}} = 4\,\text{kHz}$;
- $f_{\text{load}} = 15\,\text{Hz}$.

0.1.3 IGBTs behaviour under new performance requirements: simulation results

Comparison between **DP650B1700** and **FF1200R17IP5P** under the same working condition.

- $I_{\text{load}}^{\text{inv}} = 370\,\text{A}$, 100% of the nominal inverter current;
- $f_{\text{pwm}}^{\text{inv}} = 4\,\text{kHz}$;

- $T_{\mathrm{hs}}^{\mathrm{inv}} = 65\,^{\circ}\mathrm{C}$;
- $\cos(\varphi) = 0.8$, driver mode;
- $f_{\mathrm{load}} = 15\,\mathrm{Hz}$, load frequency (critical parameter for maximum junction temperature evaluation);
- driver board DB1.51 (new driver board).

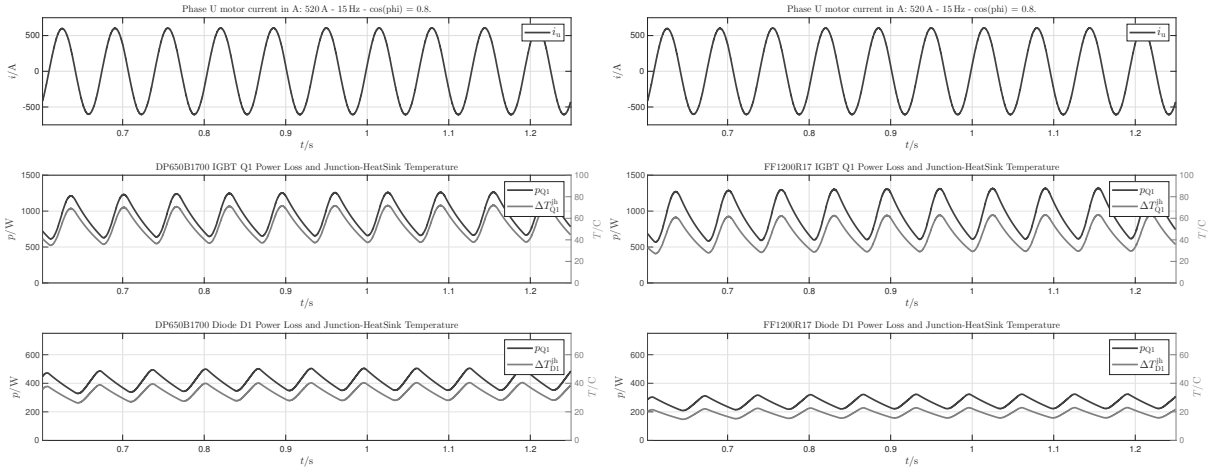


(a) DP650B1700 Q1 IGBT thermal power/thermal behaviour at $I_{\mathrm{load}}^{\mathrm{inv}} = 370\,\mathrm{A}$.

(b) FF1200R17IP5P Q1 IGBT thermal power/thermal behaviour at $I_{\mathrm{load}}^{\mathrm{inv}} = 370\,\mathrm{A}$.

Figure 1: Thermal behaviour comparison between IGBT DP650B1700 and FF1200R17IP5P at $I_{\mathrm{load}}^{\mathrm{inv}} = 370\,\mathrm{A}$.

- $I_{\mathrm{load}}^{\mathrm{inv}} = 520\,\mathrm{A}$, 140% of the nominal inverter current;
- $f_{\mathrm{pwm}}^{\mathrm{inv}} = 4\,\mathrm{kHz}$;
- $T_{\mathrm{hs}}^{\mathrm{inv}} = 65\,^{\circ}\mathrm{C}$;
- $\cos(\varphi) = 0.8$, driver mode;
- $f_{\mathrm{load}} = 15\,\mathrm{Hz}$, load frequency (critical parameter for maximum junction temperature evaluation);
- driver board DB1.51 (new driver board).



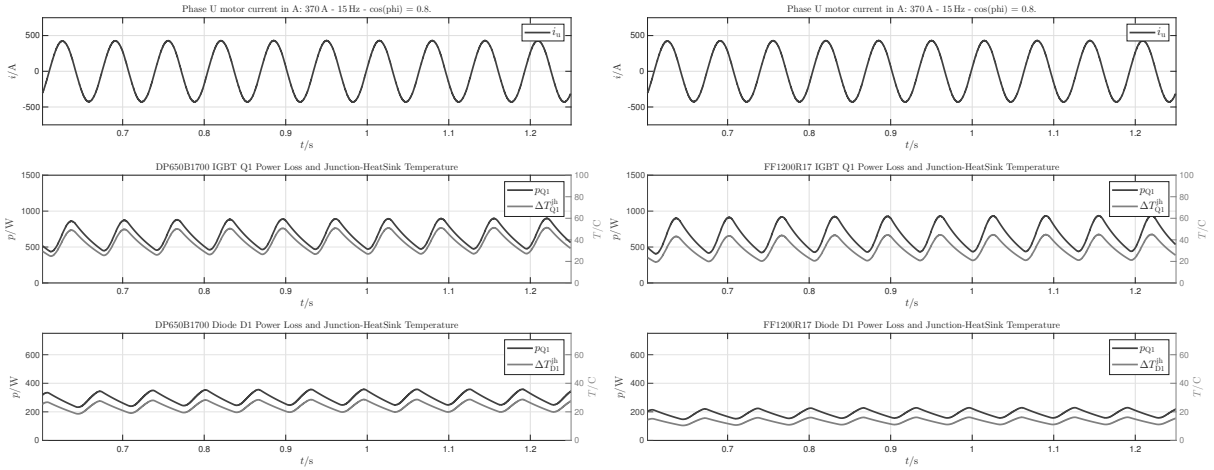
(a) DP650B1700 Q1 IGBT thermal power/thermal behaviour at $I_{\text{load}}^{\text{inv}} = 520\,\text{A}$.

(b) FF1200R17IP5P Q1 IGBT thermal power/thermal behaviour at $I_{\text{load}}^{\text{inv}} = 520\,\text{A}$.

Figure 2: Thermal behaviour comparison between IGBT DP650B1700 and FF1200R17IP5P at $I_{\text{load}}^{\text{inv}} = 520\,\text{A}$.

Comparison between **FF650R17IE4** and **FF1200R17IP5P** under the same working condition.

- $I_{\text{load}}^{\text{inv}} = 370\,\text{A}$, 100% of the nominal inverter current;
- $f_{\text{pwm}}^{\text{inv}} = 4\,\text{kHz}$;
- $T_{\text{hs}}^{\text{inv}} = 65\,^{\circ}\text{C}$;
- $\cos(\varphi) = 0.8$, driver mode;
- $f_{\text{load}} = 15\,\text{Hz}$, load frequency (critical parameter for maximum junction temperature evaluation);
- driver board DB1.51 (new driver board).

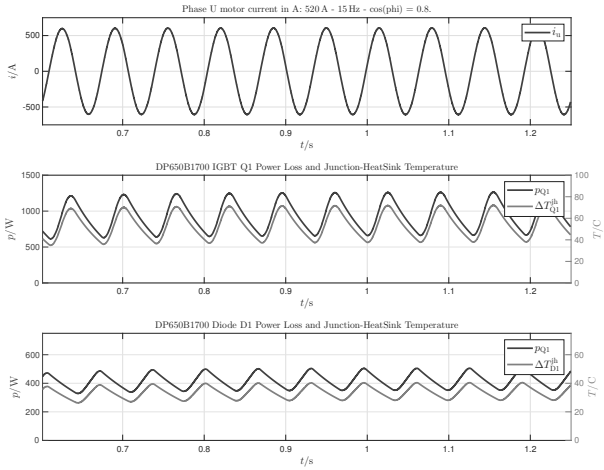


(a) FF650R17IE4 Q1 IGBT thermal power/thermal behaviour at $I_{\text{load}}^{\text{inv}} = 370\,\text{A}$.

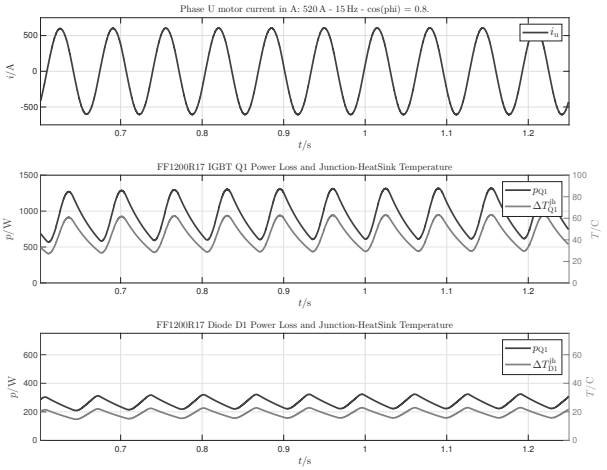
(b) FF1200R17IP5P Q1 IGBT thermal power/thermal behaviour at $I_{\text{load}}^{\text{inv}} = 370\,\text{A}$.

Figure 3: Thermal behaviour comparison between IGBT FF650R17IE4 and FF1200R17IP5P at $I_{\text{load}}^{\text{inv}} = 370\,\text{A}$.

- $I_{\text{load}}^{\text{inv}} = 520\,\text{A}$, 140% of the nominal inverter current;
- $f_{\text{pwm}}^{\text{inv}} = 4\,\text{kHz}$;
- $T_{\text{hs}}^{\text{inv}} = 65\,^{\circ}\text{C}$;
- $\cos(\varphi) = 0.8$, driver mode;
- $f_{\text{load}} = 15\,\text{Hz}$, load frequency (critical parameter for maximum junction temperature evaluation);
- driver board DB1.51 (new driver board).



(a) FF650R17IE4 Q1 IGBT thermal power/thermal behaviour at $I_{\text{load}}^{\text{inv}} = 520\,\text{A}$.



(b) FF1200R17IP5P Q1 IGBT thermal power/thermal behaviour at $I_{\text{load}}^{\text{inv}} = 520\,\text{A}$.

Figure 4: Thermal behaviour comparison between IGBT FF650R17IE4 and FF1200R17IP5P at $I_{\text{load}}^{\text{inv}} = 520\,\text{A}$.

0.1.3.1 Simulation results in tables

Simulation results case scenario 1

- $I_{\text{load}}^{\text{inv}} = 370\,\text{A}$, 100% of the nominal inverter current;
- $f_{\text{pwm}}^{\text{inv}} = 4\,\text{kHz}$;
- $T_{\text{hs}}^{\text{inv}} = 65\,^{\circ}\text{C}$;
- $\cos(\varphi) = 0.8$, drive mode;
- $f_{\text{load}} = 15\,\text{Hz}$, load frequency (critical parameter for maximum junction temperature evaluation);

Table 11: **Simulation results case scenario 1** — Power losses and maximum junction temperature at $I_{\text{load}}^{\text{inv}} = 370\,\text{A}$, $f_{\text{pwm}}^{\text{inv}} = 4\,\text{kHz}$, $T_{\text{hs}}^{\text{inv}} = 65\,^{\circ}\text{C}$, $\cos(\varphi) = 0.8$.

	T_{j}^{Q}	T_{j}^{D}	P_{Q}	P_{D}	ΔT_{j}
FF650R17IE4	120 °C	70 °C	760 W	200 W	30 °C
D650B1700	112 °C	74 °C	666 W	297 W	26 °C
FF1200R17IP5P	106 °C	61 °C	665 W	189 W	17 °C

Simulation results case scenario 2

- $I_{\text{load}}^{\text{inv}} = 520\,\text{A}$, 140% of the nominal inverter current;
- $f_{\text{pwm}}^{\text{inv}} = 4\,\text{kHz}$;
- $T_{\text{hs}}^{\text{inv}} = 65\,^{\circ}\text{C}$;
- $\cos(\varphi) = 0.8$, drive mode;
- $f_{\text{load}} = 15\,\text{Hz}$, load frequency (critical parameter for maximum junction temperature evaluation);

Table 12: **Simulation results case scenario 2** — Power losses and maximum junction temperature at $I_{\text{load}}^{\text{inv}} = 520\,\text{A}$, $f_{\text{pwm}}^{\text{inv}} = 4\,\text{kHz}$, $T_{\text{hs}}^{\text{inv}} = 65\,^{\circ}\text{C}$, $\cos(\varphi) = 0.8$

	T_{j}^{Q}	T_{j}^{D}	P_{Q}	P_{D}	ΔT_{j}
FF650R17IE4	152 °C	82 °C	1066 W	276 W	42 °C
FF650R17IE4DB2	141 °C	88 °C	940 W	417 W	37 °C
FF1200R17IP5P	132 °C	70 °C	937 W	266 W	23 °C

0.2 Double pulse tests

This section presents the results of Double Pulse Tests (DPTs) conducted to assess the performance of the FF1200R17IP5P IGBT module in conjunction with the new driver board DB1.51. The following test configurations were evaluated:

- **DPT 1**: DP650B1700 with DB1.00;
  - series-production IGBT module with series-production driver board;
- **DPT 2**: DP650B1700 with DB1.51;
  - series-production IGBT module with candidate driver board;
- **DPT 3**: FF1200R17IP5P with DB1.00;
  - candidate IGBT module with series-production driver board;

– **DPT 4**: FF1200R17IP5P with DB1.51;

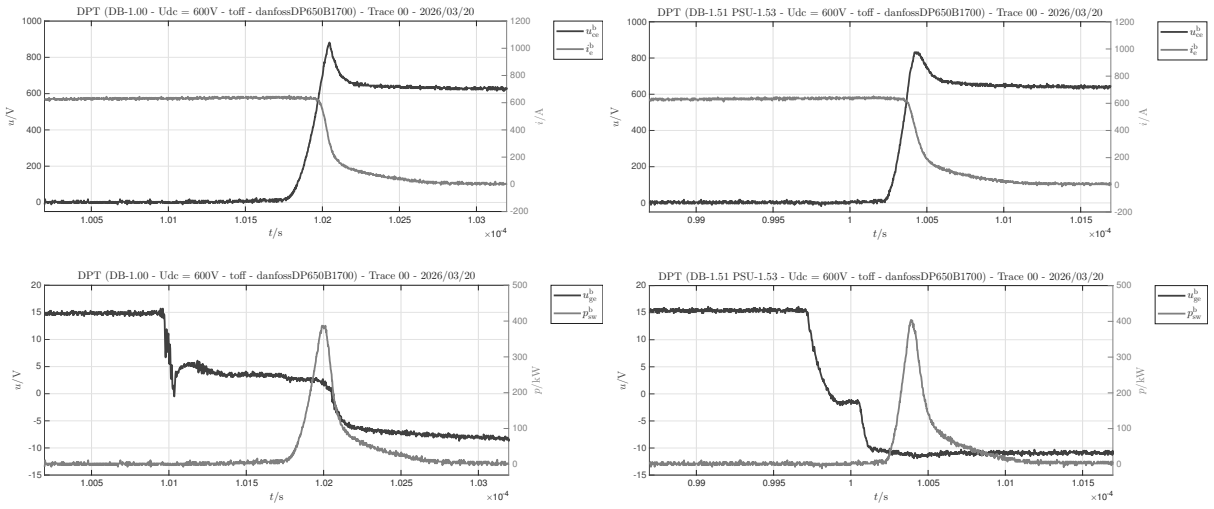
– candidate IGBT module with candidate driver board.

0.3 Measurement Results

0.3.1 Turn-OFF

The following waveform were acquired at the first turn-off command.

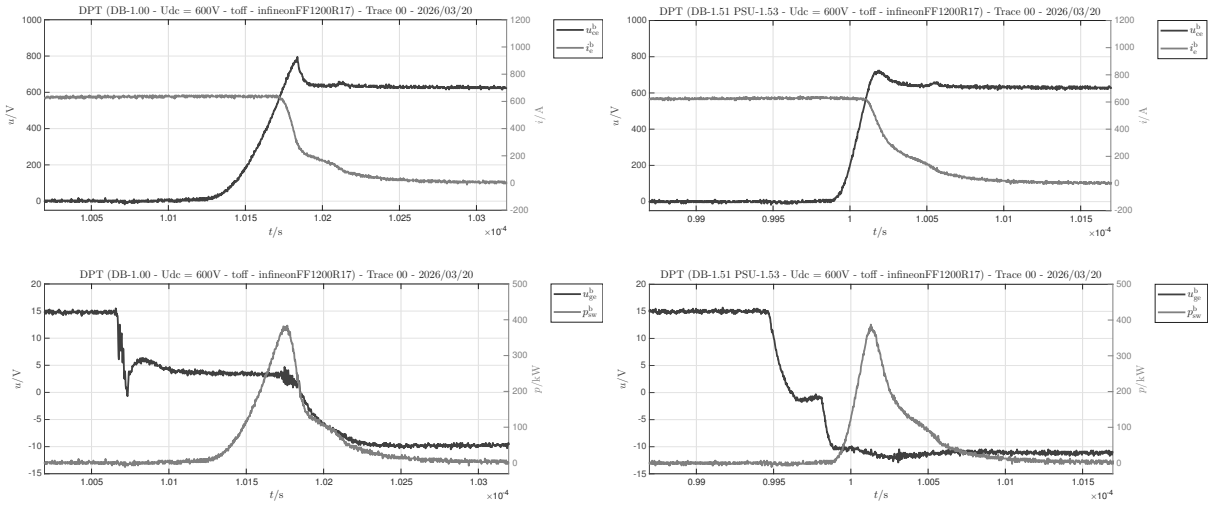
Remark: DB1.00 mounts higher turn-off resistors ($R_{\mathrm{g}}^{\mathrm{off}}$) than DB1.51, but DB1.51 implements a gate-emitter voltage rate limiter which led IGBTs to a lower turn-off collector-emitter overshoot (Δu_{ce}), as depicted in Figure 5.



(a) Turn-OFF: DP650B1700 with DB1.00.

(b) Turn-OFF: DP650B1700 with DB1.51.

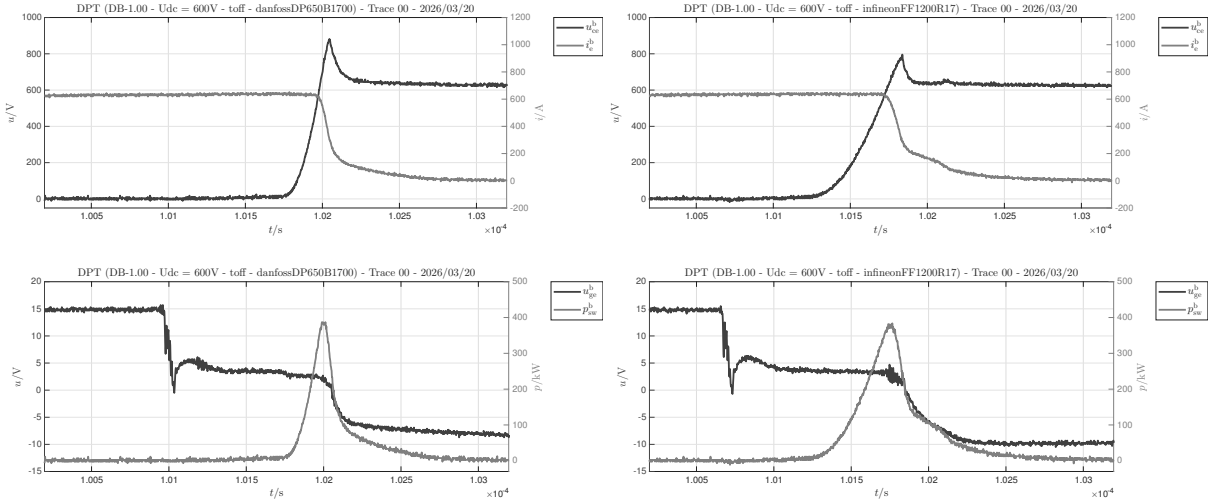
Figure 5: Turn-OFF comparison: DP650B1700 with DB1.00 and DB1.51.



(a) Turn-OFF: FF1200R17IP5P with DB1.00.

(b) Turn-OFF: FF1200R17IP5P with DB1.51.

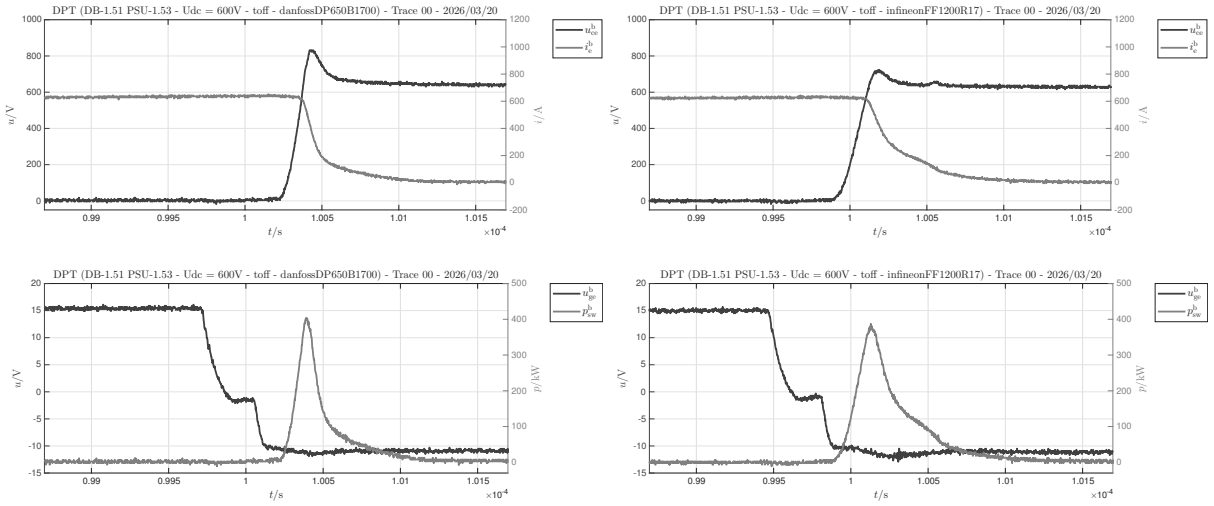
Figure 6: Turn-OFF comparison: FF1200R17IP5P with DB1.00 and DB1.51.



(a) Turn-OFF: DP650B1700 with DB1.00.

(b) Turn-OFF: FF1200R17IP5P with DB1.00.

Figure 7: Turn-OFF comparison: DP650B1700 and FF1200R17IP5P with DB1.00.



(a) Turn-OFF: DP650B1700 with DB1.51.

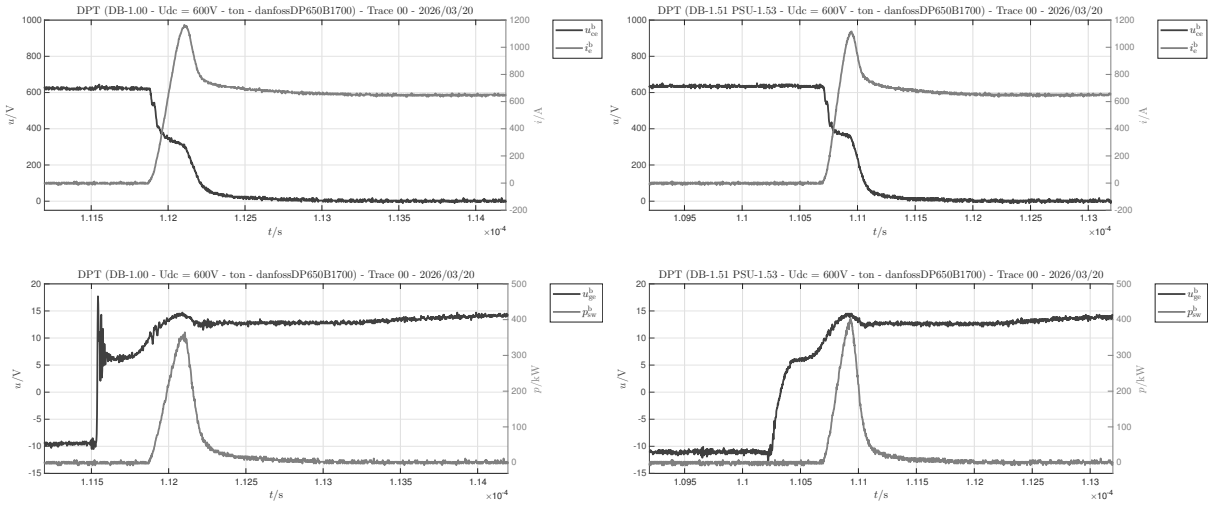
(b) Turn-OFF: FF1200R17IP5P with DB1.51.

Figure 8: Turn-OFF comparison: DP650B1700 and FF1200R17IP5P with DB1.51.

0.3.2 Turn-ON

The following waveform were acquired at the second turn-on command.

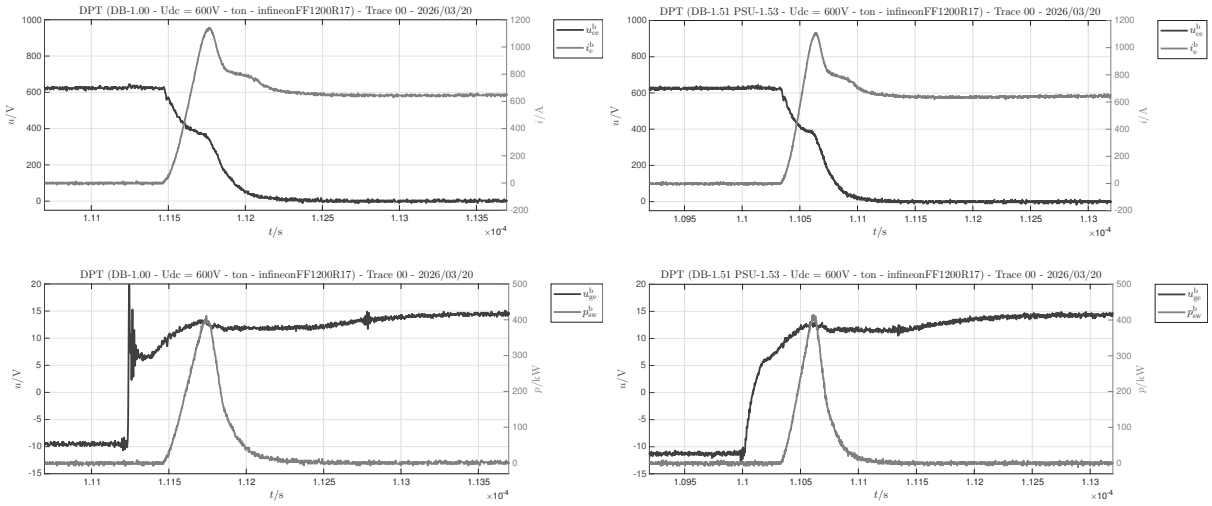
Remark: DB1.00 mounts similar turn-on resistors ($R_{\mathrm{g}}^{\mathrm{on}}$) than DB1.51, but IGBTs turn-on with DB1.51 show a reverse recovery current (I_{rr}) slightly lower due to the presence of gate command rate limiter.



(a) Turn-ON: DP650B1700 with DB1.00.

(b) Turn-ON: DP650B1700 with DB1.51.

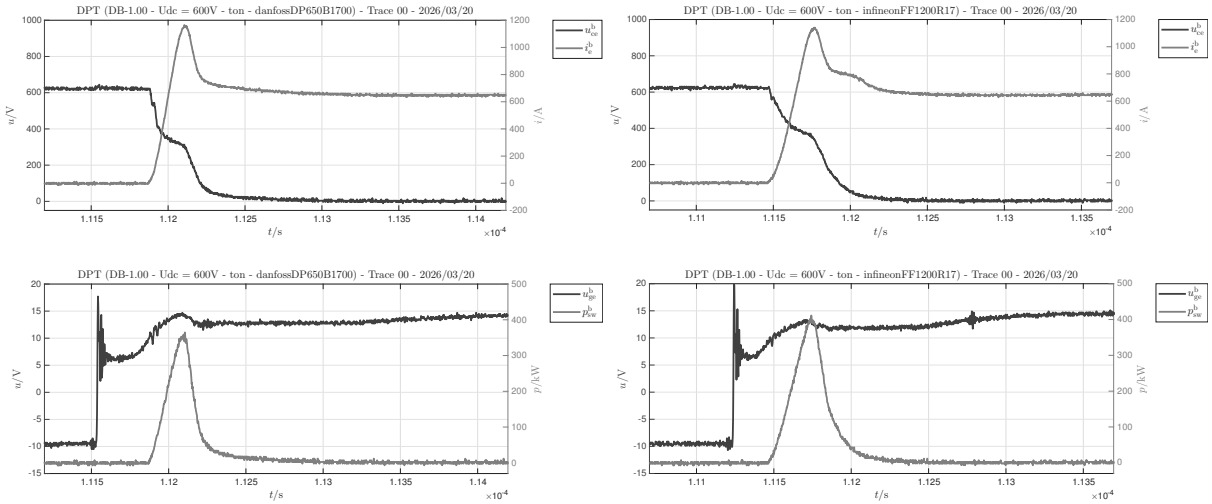
Figure 9: Turn-ON comparison: DP650B1700 with DB1.00 and DB1.51.



(a) Turn-ON: FF1200R17IP5P with DB1.00.

(b) Turn-ON: FF1200R17IP5P with DB1.51.

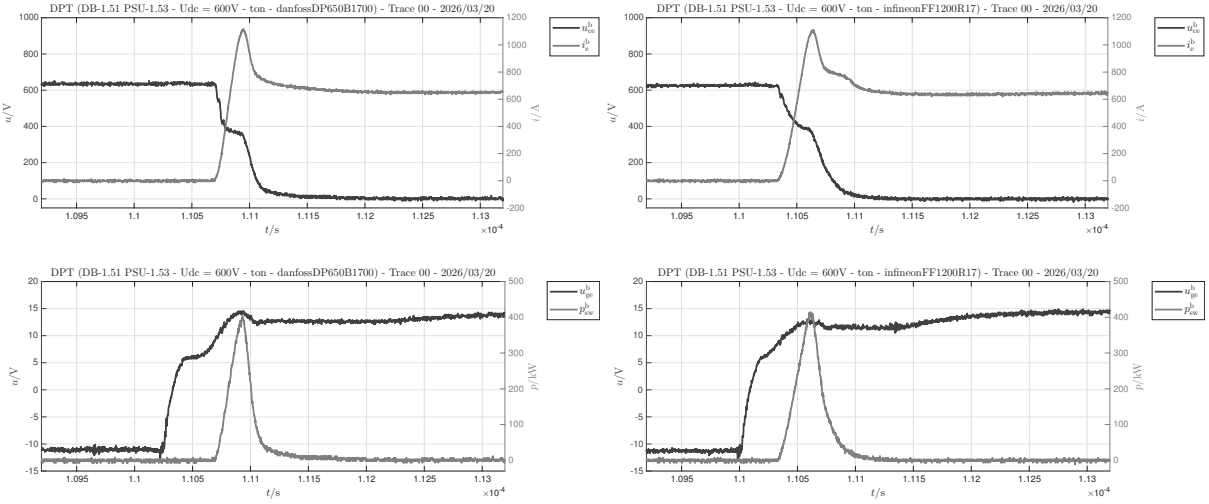
Figure 10: Turn-ON comparison: FF1200R17IP5P with DB1.00 and DB1.51.



(a) Turn-ON: DP650B1700 with DB1.00.

(b) Turn-ON: FF1200R17IP5P with DB1.00.

Figure 11: Turn-ON comparison: DP650B1700 and FF1200R17IP5P with DB1.00.



(a) Turn-ON: DP650B1700 with DB1.51.

(b) Turn-ON: FF1200R17IP5P with DB1.51.

Figure 12: Turn-ON comparison: DP650B1700 and FF1200R17IP5P with DB1.51.

0.3.3 Summary of Test Results

The tests were carried out under the four configurations defined in Section 0.2, repeated here for convenience:

- **Case 1**: DP650B1700 with DB1.00;
  - series-production IGBT module with series-production driver board;
- **Case 2**: DP650B1700 with DB1.51;
  - series-production IGBT module with candidate driver board;
- **Case 3**: FF1200R17IP5P with DB1.00;
  - candidate IGBT module with series-production driver board;
- **Case 4**: FF1200R17IP5P with DB1.51;
  - candidate IGBT module with candidate driver board.

The main results are summarised in Table 13.

Table 13: Summary of DPT results.

	Case 1	**Case 2**	**Case 3**	**Case 4**
E_{on} [mJ]	80	91	105	107
E_{off} [mJ]	90	93	152	130
I_{rr} [A]	512	448	488	464
Δu_{ce} [V]	245	187	173	86

Remark — The gate resistances mounted on the two driver boards are as follows:

$$\begin{aligned} R_{\text{g}}^{\text{on}}\Big|_{\text{DB1.00}} &= \frac{6.2\,\Omega}{5} = 1.24\,\Omega, \qquad R_{\text{g}}^{\text{off}}\Big|_{\text{DB1.00}} = \frac{18.0\,\Omega}{5} = 3.6\,\Omega \\ R_{\text{g}}^{\text{on}}\Big|_{\text{DB1.51}} &= \frac{4.47\,\Omega}{4} = 1.175\,\Omega, \qquad R_{\text{g}}^{\text{off}}\Big|_{\text{DB1.51}} = \frac{4.47\,\Omega}{4} = 1.175\,\Omega \end{aligned} \tag{0.3.1}$$

The reference data shown in Table 14 have been scaled from the device datasheets, accounting for collector current I_c, collector-emitter voltage V_{ce}, and case temperature T_{case} (assumed equal to the junction temperature) using simple linear scaling.

DPTs were carried out at $I_c = 620\,A$ and $V_{ce} = 630\,V$, yielding the following reference switching energy values:

Table 14: Reference E_{on} and E_{off} for the DPTs.

	T_{case}	E_{on}^{ref}	E_{off}^{ref}	I_c	V_{ce}	R_g^{on}	R_g^{off}
FF650R17IE4	25 °C	136 mJ	93 mJ	620 A	630 V	1.24 Ω	3.6 Ω
FF1200R17IP5P	25 °C	132 mJ	100 mJ	620 A	630 V	1.175 Ω	1.175 Ω

Table 15: Measured E_{on} and E_{off} with DB1.51.

	T_{case}	E_{on}^{m}	E_{off}^{m}	I_c	V_{ce}	R_g^{on}	R_g^{off}
FF650R17IE4	25 °C	91 mJ	93 mJ	620 A	630 V	1.175 Ω	1.175 Ω
FF1200R17IP5P	25 °C	107 mJ	130 mJ	620 A	630 V	1.175 Ω	1.175 Ω

Table 16: Measured E_{on} and E_{off} with DB1.00.

	T_{case}	E_{on}^{m}	E_{off}^{m}	I_c	V_{ce}	R_g^{on}	R_g^{off}
FF650R17IE4	25 °C	80 mJ	90 mJ	620 A	630 V	1.24 Ω	3.6 Ω
FF1200R17IP5P	25 °C	105 mJ	152 mJ	620 A	630 V	1.24 Ω	3.6 Ω

Remark: The new IGBT module FF1200R17IP5P exhibits higher switching power losses compared to the other modules under the same load conditions; however, it demonstrates superior performance in terms of conduction power losses. Furthermore, the FF1200R17IP5P features a lower thermal impedance, resulting in more efficient thermal energy dissipation. As a consequence, the maximum junction temperature is approximately 15 °C lower than that of the other modules. It is worth noting that this device is rated for a maximum junction temperature of 175 °C, whereas the remaining modules are rated at 150 °C, providing an additional thermal safety margin.



0.4 Overall Results and Conclusions

Based on the data collected in the previous sections, the FF1200R17IP5P module in combination with the DB1.51 driver board demonstrates full compatibility and satisfies all design requirements. It is therefore recommended to proceed with further investigations, including load testing, to validate the module under real operating conditions.

Bibliography

[1] D. A. Neamen *Semiconductor Physiscs and Devices*. Springer-Verlag. McGraw Hill, Fourth Edition 2012.